

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	MANOJ M	Reg. No.:	192511060
Section / Batch:	CSE	Date:	29.07.2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Q51. A smart grid system collects power consumption readings that arrive in random order and must be sorted periodically for analysis.

- Analyze how Selection Sort and Bubble Sort perform on this dataset.
- Compare their efficiency in terms of comparisons and swaps.
- Recommend the most suitable algorithm with justification

Code of Conduct

I Manoj M(192511060) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				

Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

a) Execution of Selection Sort and Bubble Sort

Given Array: 29, 10, 14, 37, 13

Selection Sort

- **Pass 1:** Minimum = 10 → Swap with 29
29 14 37 13
- **Pass 2:** Minimum = 13 → Swap with 29
10 13 14 37 29
- **Pass 3:** Minimum = 14 → No Swap
10 13 14 37 29
- **Pass 4:** Minimum = 29 → Swap with 37
10 13 14 29 37

Final Sorted Array: 10 13 14 29 37

Bubble Sort

- **Pass 1**
 - 29 & 10 → Swap → 10 29 14 37 13
 - 29 & 14 → Swap → 10 14 29 37 13
 - 29 & 37 → No Swap
 - 37 & 13 → Swap → 10 14 29 13 37
 - **Pass 2**
 - 10 & 14 → No Swap
 - 14 & 29 → No Swap
 - 29 & 13 → Swap → 10 14 13 29 37
 - **Pass 3**
 - 10 & 14 → No Swap
 - 14 & 13 → Swap → 10 13 14 29 37
 - **Pass 4**
 - 10 & 13 → No Swap → Array Sorted
- Final Sorted Array:** 10 13 14 29 37
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b) Comparison

Criteria	Selection Sort	Bubble Sort
Comparisons	10	10
Swaps	3	4
Time Complexity	$O(n^2)$	$O(n^2)$

c) Recommendation

Selection Sort is the better choice because it performs **fewer swaps (3)** while making the same number of comparisons (**10**) as Bubble Sort. Since power consumption data is sorted periodically, minimizing swaps reduces data movement and improves efficiency.